

Form number	Form name	Enrollment year	Fill in moment
1	CSE Program of Examinations	2021/2022 or later	At least 6 weeks before start preparation project

Name:

ID Number:

Intended graduation cluster:

Month and year of enrollment:

Name representative research cluster:

Instructions

Please fill in the form digitally. In the case you want to change your program and you require permission from the Examination Committee in advance (e.g. when following courses at another university), note the changes on page five. When you want to change the program for other reasons, please hand in a revised form at the start of your preparation phase together with form 2. For more information on the CSE program check [the online education guide](#).

1. In the red column you need to select three foundational courses. You can only pick one foundational course per focus area.
2. In the blue column you need to select three extra courses from your chosen focus area. These can be either foundational courses or deepening courses. Please indicate which focus area you picked by ticking the box *focus area*.
3. Next you need to select specialization electives. You need select 30 credits of specialization electives, which can come from the green column or from the list of specialization electives on page 3.
4. Additionally, on page 3 you need to fill in your free electives (15 credits). Do you wish to include courses from another university? Please provide links to the course descriptions of these courses (e.g. a link to a course catalogue).
5. On page 4 you can fill in your homologation courses if applicable and choose the seminar you wish to follow.
6. If you need to make changes to a previously approved program please indicate the changes made in the text box on page 5 as well.

Focus areas				
Algorhitms and Theory		Focus area		
Code	Course title	Foundational courses	Extra courses	Specialization electives
2IMA10	Advanced Algorithms			
2IMF25	Automated Reasoning			
2IMF10	Process Algebra			
2IMA20	Algorithms for Geovisualization			
2IMA15	Geometric Algorithms			
2IMA35	Massively Parallel Algorithms			
2IMA25	Exact Algorithms for NP-hard Problems			
2IMF15	Proving with Computer Assistance			
2IMA30	Topological Data Analysis			
2IMA50	Algorithms for Collective Decision Making			

Focus areas				
Architectures and Systems		Focus area		
Code	Course title	Foundational courses	Extra courses	Specialization electives
2IMN10	Architecture of Distributed Systems			
2IMF30	System Validation			
2IMD10	Engineering Data-Intensive Systems			
2IMF35	Algorithms for Model Checking			
2IMN15	Internet of Things			
2IMN20	Real-Time Systems			
2IMN25	Quantitative Evaluation of Cyber Physical Systems			
2IMN35	VLSI programming			
2IMN30	Machine Learning for Industry			

Focus areas				
Software and Analytics		Focus area		
Code	Course title	Foundational courses	Extra courses	Specialization electives
2IMP25	Software Evolution			
2AMI10	Foundations of Process Mining			
2IMP10	Program Verification Techniques			
2AMM10	Deep Learning			
2AMM20	Research Topics in Data Mining*			
2IMN30	Machine Learning for Industry			
2IMP40	Empirical Methods in Software Engineering			
2AMD15	Big Data Management			
2IMP30	System Design Engineering			
2IMP20	Domain Specific Language Design			
2IMP60	Human-Computer Interaction			
Subtotal credits		15	15	

* Only students that started their Program before AY 2023-2024 may take this course as foundational course in the Software and Analytics Focus Area

Specialization electives		
Code	Course title	
2IMS10	Physical Aspects of Digital Security	
2IMS25	Principles of Data Protection	
2IMV25	Interactive virtual environments	
2DMI20	Software Security	
2IMS20	Cyberattacks Crime and Defenses	
2AMM15	Machine Learning Engineering	
2IMD20	Language Virtual Machines Design and Implementation	
2IMS15	Verification of Security Protocols	
2IMS30	Advanced Network Security	
2IMS40	Intrusion Detection Laboratory	
2IMS50	Introduction to Quantum Computing and Security	
2IMS60	Provable Security	
2IMV10	Visual Computing Project	
2AMM10	Deep Learning	
2IMP15	Software Project Management	
2IMV15	Simulation in Computer Graphics	
2IMC10	Internship	
Subtotal credits		

Internship supervisor (if known):

An internship is optional. Keep in mind, when you do an external internship (e.g. at a company), your graduation project needs to be executed internally (within TU/e), when you do an internal internship you cannot graduate with the same supervisor.

Free elective courses		
Code	Course title	Credits
Subtotal credits		

Homologation courses (if applicable)(homologation courses count towards the 15 credits in free elective courses)**

**Homologation courses are bachelor courses assigned during the admission process to make up deficiencies in previous knowledge. Please check your admission letter to see if you have homologation courses. It is also possible to pick a maximum of three bachelor courses yourself to compensate deficiencies, if you think it is necessary. If you do that, a motivation for including the self-chosen homologation courses must be attached to this form.

Homologation courses		
Code	Course title	Credits
Subtotal credits		

Seminar		
Code	Course title	Credits
2IMD00	Seminar Datamanagement	
2IMF00	Seminar Formal System Analysis	
2IMI00	Seminar Process Analytics	
2IMM00	Seminar Data Mining	
2IMN00	Seminar Interconnected Resource-aware Intelligent Systems (IRIS)	
2IMP00	Seminar Software Engineering and Technology	
2IMU00	Seminar Uncertainty in AI	
2IMV00	Seminar Visualization	
2IMA00	Seminar Algorithms	
2IMS00	Seminar Information Security Technology	
Subtotal credits		

Graduation Project		
Code	Course title	Credits
2IMC05	Preparation Graduation Project	10
2IMC00	Master Project	30
Subtotal credits		40

Total number of credits (at least 120 credits)	
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Agreement student

Signature student:

Date:

Changes to the previously approved program, links to course descriptions of courses followed at another university (including a motivation for your selection and an explanation of why these courses do not overlap with each other or with other courses in your program), and/or motivation for self- chosen homologation courses (if applicable):

If you do not agree with the decision of the Examination Committee, you may submit an appeal via this webpage within a period of six weeks after the date of this decision.

This section to be completed by the Examination Committee

Approval Examination Committee:

Date: